CHD

Saturday, January 8, 2022 2:26 PM

❖ Overview

➢ **Incidence**

- VSD (30%)
- PDA (12%)
- ASD (7%)

➢ **Risk Factors**

- **family history of CHD**
- **Exposures during pregnancy**
  - Drugs e.g. warfarin, anticonvulsants, alcohol
  - Diseases e.g. maternal rubella (PDA), Maternal Diabetes Mellitus
- **Chromosomal**
  - Down syndrome → ASD, VSD
  - Turner syndrome → Aortic stenosis, coarctation of aorta
  - Williams syndrome → Supra valvular aortic stenosis , pulmonary stenosis

➢ **Presentation**

- incidental murmurs
- Manifestations of ↑ pulmonary blood flow
  - Poor feeding
  - exercise intolerance
  - Recurrent chest infections
  - Recurrent heart failure
  - Wheezes
  - Growth failure
- **Complications**
  - Congestive heart failure
  - infective endocarditis
  - Growth retardation
  - Thromboembolism
  - Polycythemia

▪ **Classification of CHD**

- Acyanotic Congenital Heart Diseases (ACHD)
- Cyanotic Congenital Heart Disease (CCHD)

❖ **Acyanotic Congenital Heart Diseases (ACHD)**

○ constitutes 80% of all CHD

➢ **Classification**

○ **ACHD with left to right shunt**

- (VSD)
- (PDA)
- (ASD)

○ **ACHD without shunt**

- **Obstructive lesions**
  - Aortic coarctation
  - Aortic stenosis
  - Pulmonary stenosis
- **Non obstructive lesions**
  - Dextrocardia
  - Mitral valve prolapse

❖ Ventricular septal defect (VSD)

➢ **Types of VSDs**

- Perimembraneous defect (70%)
- Muscular defects
- Outlet defect (infundibular or sub arterial)

➢ **Clinical features**

- Manifestations of ↑ pulmonary blood flow
- Breathlessness > poor feeding
- tachypnea
- Tachycardia
- enlarged tender liver
- Murmur of VSD
- Accentuated P2

➢ **Investigations**

○ **Chest X-ray**

- Cardiomegaly
- plethoric lungs
- Pulmonary edema

○ ECG > signs of BVH

- Echo > the defect & its size , EF

➢ **Complications**

- Infective endocarditis
- Heart failure
- Higher risk of **Eisenmenger syndrome**
- Stunted growth
- Recurrent chest infection

➢ **Management**

○ **Small defect**

- Conservative Mx
- Avoid infective endocarditis by dental hygiene and antibiotic prophylaxis.
- Follow up > by ECG & Echo every 6 month to confirm spontaneous closure

○ **Large defect**

- **Medical**
  - Control heart failure (Diuretics, Captopril)
  - Antibiotics for chest infections
  - Additional calorie input & monitor growth
  - Avoid infective endocarditis
  - Follow up with ECG & Echo to confirm spontaneous closure
- **Surgical**
  - Types
    - Palliative: Pulmonary artery banding > ↓ PBF
    - Direct closure of the defect
  - Indications
  - Less than 6 months
    - Symptomatic large defects with uncontrollable heart failure or Growth failure
    - Progressive pulmonary hypertension
  - After 1st yr if a QP:QS ratio > than or equal to 2:1

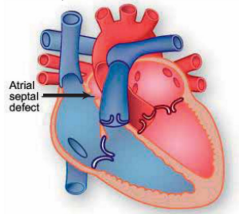
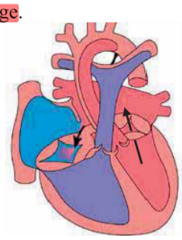
➢ **Prognosis**

○ 30-50% of small defects (especially muscular) close spontaneously within the first 2-years of life

❖ Atrial Septal Defect (ASD)

➢ **Types of ASD**

- Secundum ASD 80%
- Primum ASD (Partial AVSD)
- Sinus venosus defect

Secundum ASD	Primum ASD
The commonest type (80% of ASD) Lies in the middle part of the septum Association May be with **Holt Oram** syndrome (Absent radii, 1st degree heart block, secundum ASD). 	• Less common • Lies in the lower part of the septum • Association Usually associated with cleft of mitral valve leaflet→ mitral regurge. 

➢ **C/F**

- Asymptomatic > Small lesion
- features of increased pulmonary blood flow
- Arrhythmias (4th decade onwards)
- Cardiomegaly with BVH
- Murmur of a relative pulmonary stenosis
  - Soft ejection systolic without thrill
  - accentuated P2
- Wide fixed splitting of S2
- Pansystolic murmur of mitral regurge
- tricuspid mid-diastolic murmur

➢ **Complications**

- Recurrent heart failure
- Recurrent pulmonary infections
- Infective endocarditis
- Eisenmenger syndrome

➢ **Investigations**

○ **Chest X-ray**

- Cardiomegaly > BVH & RAD
- Plethoric lungs

○ **ECG**

- RBBB
- right axis deviation

○ Echocardiography

○ Cardiac catheter

➢ **Management**

○ **Medical**

- Same lines as for VSD
- Infective endocarditis prophylaxis is needed only for primum ASD with mitral regurge

○ Surgical or transcatheter device closure

- symptomatic patients
- Asymptomatic patients with a QP : QS ratio of at least 2: 1
- RVH

❖ Patent ductus arteriosus (PDA) راجع من الاستب سؤال 26

○ Commonest CHD in neonates

○ Mostly affects preterms

○ Normally close within 3 month after birth

➢ **Association**

- Congenital rubella syndrome
- prematures

- **Clinical features**
  - Asymptomatic
  - Hyperdynamic circulation → collapse pulse
  - ↑ Pulse pressure, bounding pulse
  - Symptoms of ↑ pulmonary blood flow
  - Evidence of LVH
  - Continuous machinery murmur (Gibson murmur)
- **Investigations**
- **Chest X ray**
  - Cardiomegaly
  - Plethoric lungs
- ECG
  - LVH & RVH
- Echocardiography
- **Complications**
  - As in large VSD
  - Aneurismal dilatation
  - rupture of the duct
- **Management**
- **Medical**
  - Control heart failure
  - prevent infective endocarditis
  - Medical closure in preterm by I.V. **indomethacin** in the 1st week of life
- **Transcatheter or Surgical closure**
  - all PDAs should be closed, (before 1 yr of age)
  - Take care of left recurrent laryngeal nerve ⚠
- **DDx of continuous murmurs**
- aorticopulmonary window defect
- A sinus of Valsalva aneurysm that has ruptured into the right side of the heart or pulmonary artery, Coronary arteriovenous fistulas
- Truncus arteriosus
- VSD with aortic insufficiency, and combined aortic and mitral insufficiency (to-and-fro murmur rather than continuous nature)

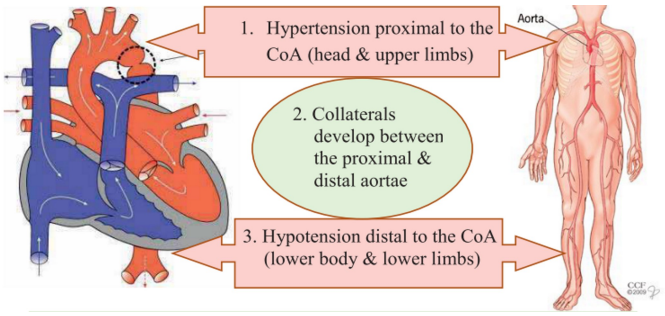
❖ Atrio ventricular septal defect (AVSD)

- Large ASD & VSD
- incompetent atrio ventricular valve

- **Association**
  - Down syndrome
- **Clinical features**
  - very similar to large VSD and cases are only diagnosable by Echocardiography
  - Features of ↑ pulmonary blood flow
  - intractable heart failure
  - Evidence of cardiomegaly
  - hyperactive precordium
  - Systolic thrill on lower left sternal border
  - Pansystolic murmur over Lower left sternal border
  - Pulmonary area → Systolic murmur
- **Treatment**
  - Medical: As for large VSD
  - Surgical
    - Early repair > to avoid pHTN & intractable HF
    - Pulmonary artery banding
- **Complication**
- surgically induced heart block

❖ Coarctation of Aorta (CoA)

- **Types**
  - Ductal, Preductal or postductal
- **Association**
  - Turner syndrome
  - Bicuspid aortic valve



- **Clinical features**
  - Asymptomatic
  - poor feeding
  - congestive heart failure
  - Cardiomegaly
  - Acidosis
  - **Pulse**
    - Bounding in upper limbs and carotids , weak or absent in lower limbs
    - Radio-femoral delay
  - **Blood pressure**
    - Higher in upper limbs than lower limb
    - Differ by 20 mmHg in systolic
  - **Murmur**
    - The Only murmur herd from posterior
    - Interscapular Systolic murmur with ejection click
- **Investigations**
  - **Chest X-ray**
    - LVH
    - Notching of the inferior border of the ribs > **Rosler sign**) > due to collateral vessels
  - Echo & aortogram
- **Complications**
  - Systemic hypertension > due to renal hypoperfusion
  - Intracranial hemorrhage > due to HTN or associated aneurysm of circle of Willis
- **Management**
  - **In neonates with severe coarctation of the aorta**
    - Infusion of prostaglandin E1
    - surgical repair
  - **Older cases**
    - Control of HTN
    - infective endocarditis prophylaxis
    - surgical repair

- **cyanotic Congenital Heart Diseases (CCHD)**

Classification

CCHD with decreased pulmonary blood flow	CCHD with increased pulmonary blood flow
With RVH - Fallot tetralogy - Pulmonary atresia with VSD **With LVH** - Tricusped atresia - Ebstein anomaly - Pulmonary atresia with VSD	**With RVH** - Transposition of great arteries (TGA) - Total anomalous pulmonary venous return - Eisenmenger Syndrome **With BVH** - Truncus arteriosus - Single ventricle
Criteria - Hypercyanotic spells (attacks of increasing cyanosis) - Heart failure is very rare - P_2 (pulmonary component of S_2) → ↓ - Chest X-ray → lung oligaemia	- Poor feeding "dyspnea on suckling". - Recurrent chest infections is common - Recurrent heart failure is common - P_2 → ↑ - Chest X-ray → plethora.
- Growth retardation occurs in long standing, symptomatic, uncorrected lesions.	

Central cyanosis

Definition

Bluish discoloration of skin and mucous membranes due to presence of > 5 gm /ml reduced hemoglobin in the capillary blood.

Causes

1. Congenital heart diseases are the main cause of chronic central cyanosis
   - The commonest cyanotic congenital heart disease is Fallot tetralogy.
   - The commonest cyanotic congenital heart disease presenting at birth is TGA.
2. Respiratory failure (usually acute cyanosis)

Presentation

- Early it may be overlooked and become evident only during crying and feeding
- Then it becomes apparent at rest; noted mainly in inner lips and tongue

Consequences

- Cyanotic clubbing, dusky blue skin, gray sclerae with engorged blood vessels
- Polycythemia
- Increased risk of Relative iron deficiency and Cerebral stroke

- **The commonest are 5 Ts**
  - **T**etralogy OF Fallot
  - **T**ransposition of great arteries (TGA)
  - **T**runcus arteriosus
  - **T**otal anomalous pulmonary venous return
  - **T**ricuspid atresia
- Fallot Tetralogy
  - Commonest cyanotic CHD
- **Component**
  - Pulmonary stenosis
  - RVH
  - VSD > Large
  - Aorta overrides

➢ Classification

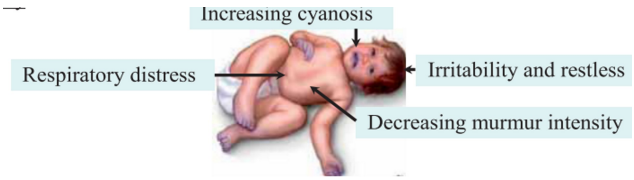
- mild degrees of PS
- Severe PS

➢ Clinical features

- Central Cyanosis
- Cyanotic Clubbing
- Growth retardation
- Squatting position
- Recurrent chest infection

○ Paroxysmal hypercyanotic Spells

- Sudden episodes of sever cyanosis & collapse
- Mainly in the 1st 2 years of life
- lasts from a few minutes to a few hours
- triggered by dehydration, crying, feeding or infection
- **Features**



- **Complications**
  - convulsions, cerebrovascular stroke or death

○ Cardiac examination

- mild RVH
- Systolic thrill over left sternal border
- S2: single (A2 only is heard)
- Murmur: systolic (organic PS) on the upper and mid left sternal border

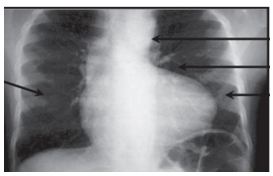
➢ Investigations

○ Chest X-ray

- Oligemic lung fields
- Rt sided aortic arch
- **Boot shaped heart**

○ Echo

- Diagnostic



○ ECG shows RVH

○ Catheter → Pre-operative

○ CBC show polycythemia

➢ Complications

○ Polycythemia

- Hypoxemia
- Relative iron deficiency

○ Cerebral thrombosis

- Extreme polycythemia
- Microcytosis

○ Brain abscess

- septic emboli
- lack of pulmonary filtration

○ Pulmonary tuberculosis

- pulmonary oligaemia

○ Infective endocarditis

○ Heart failure → rare

○ Associated anomalies

- DiGeorge syndrome { CATCH 22 }

➢ Management

○ Medical

- **Treatment of hypoxic spells**
  - hold the infant in **Knee-chest** (frog) position ± pressure to femoral pulses
  - Facial **oxygen**
  - **Morphine** 0.1mg/kg SC/IM/IV
    - Suppress the respiratory center
    - Caution in infants less than 3 months of age
  - **Fluid** bolus 10ml/kg of NaCl 0.9% or Colloid IV
  - **Na bicarbonate** 1ml/kg of 8.4% IV > if acidosis
  - **Propranolol** 0.1mg/kg IV over 5 minutes
    - Reduces infundibular spasm
  - Or **Phenylephrine** 5-10mcg/kg (slow IV push)
    - Increases systemic vascular resistance
  - PICU Intervention : Intubation and ventilation for resistant spells
- **After the spell is over**
  - Oral propranolol prophylaxis 0.5-1 mg/kg/6hours
  - Folic acid & iron supplement
  - Keep Hb conc above 12g/dl
  - Arrange for surgery
- Avoid digitalis as it may induce infundibular spasm

- **Avoid cerebral thrombosis by**
  - ttt of relative iron deficiency
  - For severe polycythemia
    - Adequate hydration
    - Phlebotomy
    - Partial exchange transfusion with albumin or saline
- Prophylaxis & treatment of infective endocarditis
- **Surgery**
- Indicated as soon as the spells begin
- Palliative shunts (**Blalock Taussig operation**)
  - Anastomosis between aorta and pulmonary artery to allow ↑ pulmonary blood flow
  - Modified : Anastomosis bet subclavian artery & ipsilateral pulmonary artery
- Total repair
  - Can be done between 4 months to 2 years according to severity and available cardiac center

➢ **DDx of TOF**
- Double outlet Rt ventricle with P.S
- Pul. Atresia With or Without VSD
- Tricuspid atresia
- Ebstein anomaly

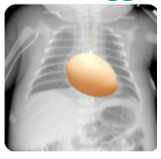
➢ **Fallot triology**
- ASD
- Pulmonary stenosis
- RVH

❖ Transposition of great arteries (TGA)

➢ **Features**
- aorta from RV & pulmonary artery from LV
- DO2 blood to systemic circ & O2 blood to the lung
- this condition incompatible with life
- Pediatric emergency > sever cyanosis at birth

➢ **C/F**
- Usually no murmurs unless associated with VSD or PS
- single second heart sound
- Symptoms of CHF

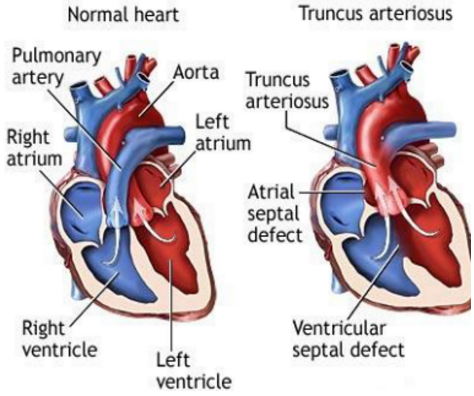
➢ **Dx > CXR**
- Plethoric lung
- RVH > Egg on side appearance




➢ **Mx**
- Medical
  - PGE1 > to keep PDA open
- Balloon atrial septostomy
  - Trans-catheter opening of ASD
- Surgical correction
  - Arterial switch or atrial switch

❖ Truncus arteriosus

➢ **Features**
- One artery trunk leave the heart
- one semilunar valve
- Large VSD below the trunk
- Genetic associations > DiGeorge syndrome



➢ **C/F**

➢ Minimal Cyanosis
- Features of ↑ pul blood flow
- Single S2
- Murmur of VSD

➢ **Dx**
- CXR
  - ↑ pulmonary vascularity
  - Rt side aortic arch
- echo

➢ **Mx**

➢ management of CHF
- Surgical correction

❖ Tricuspid atresia

➢ **Features**

- RV > hypoplastic
- ASD *must* be present to maintain blood flow
- VSD usually present

➢ **C/F**

- Sever Cyanosis at birth
- CHF

➢ **Mx**

- PGE1 > maintain PDA
- Surgical repair

❖ Total anomalous pulmonary venous return

- pulmonary veins return oxygenated blood to RA or to systemic venous system
- Pulmonary veins may be unobstructed or obstructed

➢ **C/F**

- Obstructed TAPVR > immediately after birth with profound cyanosis and respiratory distress
- Unobstructed TAPVR > increased work of breathing, tachypnea, growth failure

➢ **C/F**

- CXR
- Cardiomegaly
- snowman" shape to heart

➢ **Mx**

- Unobstructed TAPVR > surgical repair
- **Obstructed TAPVR**
  - ventilator support & supplemental oxygen
  - PGE1 infusion
  - correction of acidosis
  - emergent surgical repair

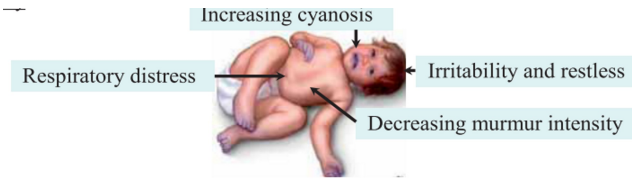
❖ Ebstein's Anomaly

➢ **Composed of**

- Downward displacement of tricuspid valve leaflets.
- Tricuspid regurge
- Huge right atrium
- Small right ventricle

➢ **Clinical features**

- May be asymptomatic
- Splitting of S1 and S2 💥
- Mild cyanosis
- Atrial arrythmias
- Pansystolic murmur (tricuspid regurge)
- heart failure

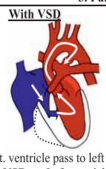
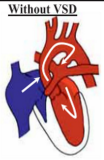
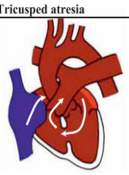


○ **Investigations**

- ECG shows RVH
- Catheter → Pre-operative
- CBC show polycythemia

➢ **Complications**

○ **Polycythemia**

3. Pulmonary atresia		4. Tricusped atresia
With VSD	Without VSD	
		
• Blood in right. ventricle pass to left ventricle via VSD so, Left ventricle contain mixed Blood → cyanosis. • Pulmonary blood flow depends on PDA **or** collaterals between Aorta & pulmonary artery.	• Blood in right atrium pass via patent foramen ovale (PFO) → left atrium → left ventricle → cyanosis. • PFO & PDA are essential for life.	• Atretic tricusped valve → Blood in right atrium pass via PFO → left atrium → left ventricle → cyanosis. • Pulmonary blood flow is dependent on VSD or PDA. • Right ventricle is hypoplastic.
Presentation		
- Cyanosis is evident at birth → increase in intensity with ductus arteriosus closure. - Single second heart sound (only A_2 is heard).		
- **No** murmurs. (may be machinery of PDA **or** collaterals).	- Murmur of VSD (± PDA).	
Diagnosis		
- Echocardiography can differentiate it from Fallot tetralogy.		
Treatment		
- Once suspected after birth: PGE_1 infusion to keep the ductus open - Operative: Palliative shunts/ Total correction - Medical: measures for polycythemia and infective endocarditis(as before)		

❖ Notes

- US Can diagnose up to 70 % of significant lesions
- VSD is the Commonest CHD & comments cause of HF in infant
- Commonest Cyanotic CHD is TOF
  TGA > 2nd most CHD & Commonest Cyanotic CHD in neonate
  ASD is the Commonest CHD in adulthood
- Eisenmenger's syndrome is only treated by lung transplant
- collapse pulse in a child think about PDA